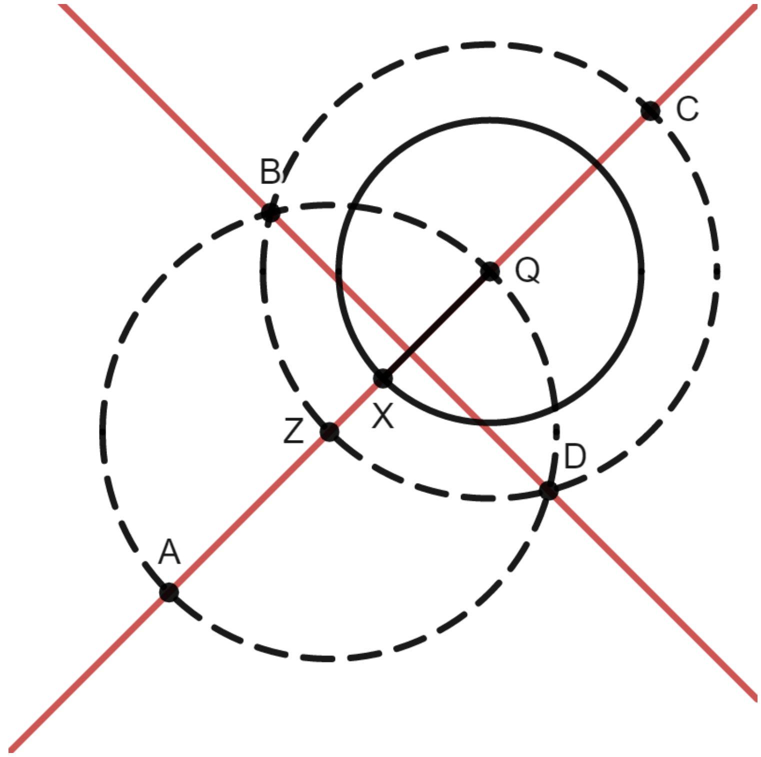
Geometry
Unit 10
Lesson H1 Practice

Name___Answer Key______
Date___________________
Period__________________



Circle Q is shown with radius $\overline{QX}$. The two dotted circles were constructed using a compass so that they are congruent to each other. Points C, B, D, and Z are on the dotted circle Q and points B, Q, D, and A on the dotted circle Z.

1. Select all of the line segments that are congruent to each other.
   - ☐ $\overline{BC}$
   - ✖ $\overline{BZ}$
   - ✖ $\overline{BQ}$
   - ✖ $\overline{QD}$
   - ✖ $\overline{QC}$
   - ✖ $\overline{QZ}$
   - ☐ $\overline{QX}$

2. Justify why the line segments chosen in question 1 are congruent.
   Answers may vary. Sample response: All line segments are the radii of both congruent circles.

3. Select all of the line segments that are congruent to each other.
   - ☐ $\overline{BD}$
   - ☐ $\overline{AB}$
   - ✖ $\overline{BZ}$
   - ☐ $\overline{AD}$
   - ✖ $\overline{AZ}$
   - ✖ $\overline{ZD}$
   - ✖ $\overline{ZQ}$
   - ☐ $\overline{ZX}$

4. Justify why the line segments chosen in question 3 are congruent.
   Answers may vary. Sample response: The line segments are radii in Circle Z.

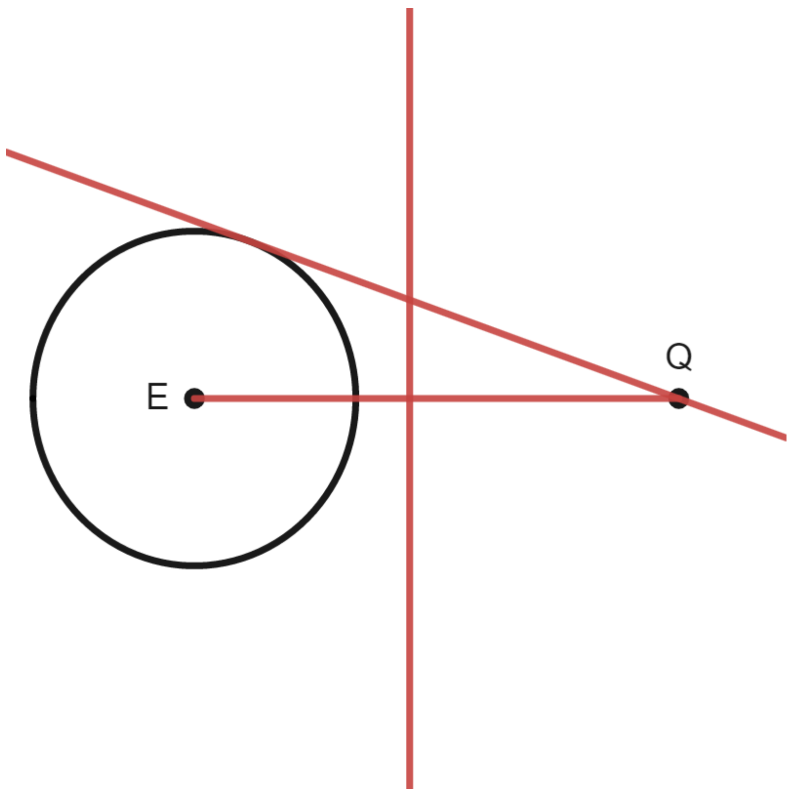
5. Draw $\overleftrightarrow{AC}$ on the diagram.
See diagram.

6. Draw $\overleftrightarrow{BD}$ on the diagram.
See diagram.

7. Describe how $\overleftrightarrow{AC}$ and $\overleftrightarrow{BD}$ are related.
$\overleftrightarrow{AC} \perp \overleftrightarrow{BD}$

8. Justify whether $\overleftrightarrow{BD}$ is tangent to circle Q.
$\overleftrightarrow{BD}$ is not tangent to Circle Q. It meets Circle Q twice. Need Point X as midpoint of $\overline{ZQ}$ by separating Circle Z and Circle Q some more.

Circle E and point Q are shown.



9. Construct the perpendicular bisector of $\overline{EQ}$.
See diagram.

10. Construct a line that is tangent to Circle E and goes through point Q.
See diagram.